Assignment 1: Multi-class Diabetes Risk Prediction

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1 Dataset and Preparation

1.1 Data Splitting and Random Seed

To ensure reproducibility, all random operations were initialized with a fixed random seed of 42. The dataset was split into 70% training, 10% validation, and 20% testing using stratified splitting to maintain the original class distribution across all subsets.

1.2 Feature Preprocessing and Selection

Numerical features, specifically BMI, MentHlth, PhysHlth, Education, Income and Age, were standardized using a standard scaler to prevent attributes with larger scales from mathematically dominating distance calculations in models like KNN.

To identify the most predictive features and reduce dimensionality, we applied the Chi-Squared (χ^2) statistical test using a SelectKBest approach. This allowed us to filter the original 21 features down to the 10 most informative attributes, streamlining model efficiency and focusing on the most statistically significant predictors of diabetes risk.

1.3 Handling Class Imbalance

The original dataset suffers from severe class imbalance. To mitigate this during training, we explored three techniques: Random Oversampling, SMOTE (Synthetic Minority Oversampling Technique), and Random Undersampling. Both Random Oversampling and SMOTE successfully balanced the training set by increasing the minority classes to match the majority class, resulting in exactly 133,038 samples per class. SMOTE yielded the best validation performance for our distance-based models.

2 Implementation Details

For each of the following models, two versions were trained: a Baseline version on the original imbalanced training set, and a Balanced version using the handled dataset. Hyperparameters were tuned exclusively using the 10% validation set.

2.1 K-Nearest Neighbors (KNN)

During the hyperparameter tuning phase on the validation set, we experimented with k values of 3, 5, 11, and 21, and evaluated both Euclidean and Manhattan distance metrics. For the Baseline model trained on imbalanced data, the optimal configuration was found to be k=5 using the Manhattan distance. For the Balanced model trained on the SMOTE dataset, the best configuration remained k=5 with Manhattan distance (though Random Oversampling favored Euclidean).

2.2 Softmax Regression

For Softmax (Multinomial Logistic) Regression, the regularization parameter C was tuned over a logarithmic grid: $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100, 10^3, 10^4\}$. The model was trained using the `lbfgs` solver with `multi_class=multinomial` and a maximum of 5000 iterations. The optimal C was selected by maximising the **Macro-F1** score on the validation set, making the selection criterion directly aligned with the primary evaluation metric for this imbalanced dataset.

Beyond the standard balancing strategies (Oversampling, SMOTE, Undersampling, SMOTEENN, and Class Weights), one additional technique was explored:

- **Per-class Threshold Tuning:** Instead of using the default 0.5 decision threshold for all classes, a grid search over thresholds $[0.1, 0.9]$ was performed on the validation set to find the per-class threshold combination that maximises Macro-F1. The best thresholds found were $[0.1, 0.3, 0.3]$ for No Diabetes, Pre-diabetes, and Diabetes respectively.

2.3 Custom Feedforward Neural Network (FNN)

3 Evaluation

3.1 Metric Justification

For this specific imbalanced dataset, **Macro-F1** is the most meaningful and reliable metric. Because the "No Diabetes" class vastly outnumbers the "Prediabetes" class, Accuracy and Micro-F1 are highly misleading; a model can achieve over 80% accuracy simply by guessing the majority class every time. Macro-F1 calculates the F1-score for each class independently and averages them equally, punishing models that completely fail to identify the minority classes.

3.2 Results Summary

Table 1 summarizes the performance of all trained models on the isolated test set.

Table 1: Model Performance Comparison on Test Set

Model Version	**Accuracy**	**Micro-F1**	**Macro-F1**	**Weighted-F1**
KNN (Baseline)	0.8166	0.8166	0.3994	0.7870
KNN (SMOTE)	0.7177	0.7177	0.4166	0.7453
KNN (Oversamp.)	0.7032	0.7032	0.4131	0.7347
KNN (Undersamp.)	0.5827	0.5827	0.3987	0.6693
Softmax (Baseline)	0.8331	0.8331	0.3886	0.7900
Softmax (Oversamp.)	0.6340	0.6340	0.4267	0.7100
Softmax (SMOTE)	0.6384	0.6384	0.4276	0.7100
Softmax (Undersamp.)	0.6893	0.6893	0.4323	0.7300
Softmax (SMOTEENN)	0.5915	0.5915	0.4142	0.6800
Softmax (Class Weights)	0.6501	0.6501	0.4310	0.7200
Softmax (Thr. Tuning)	0.7783	0.7783	0.4602	0.7900
FNN (Baseline)	0.0000	0.0000	0.0000	0.0000
FNN (SMOTE)	0.0000	0.0000	0.0000	0.0000
FNN (Oversamp.)	0.0000	0.0000	0.0000	0.0000
FNN (Undersamp.)	0.0000	0.0000	0.0000	0.0000

3.3 Confusion Matrices

3.3.1 K-Nearest Neighbors (KNN)

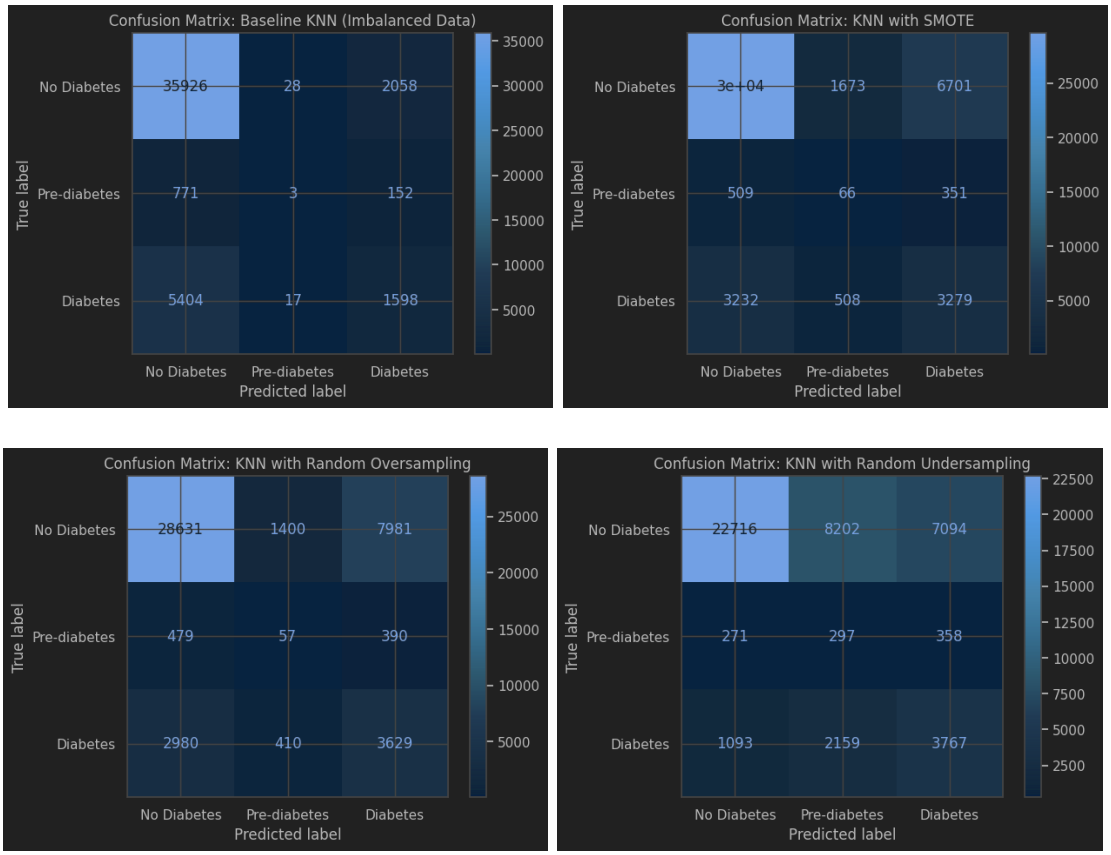

Figure 1: KNN Confusion Matrices (Top Left: Baseline, Top Right: SMOTE, Bottom Left: Oversampled, Bottom Right: Undersampled)

3.3.2 Softmax Regression

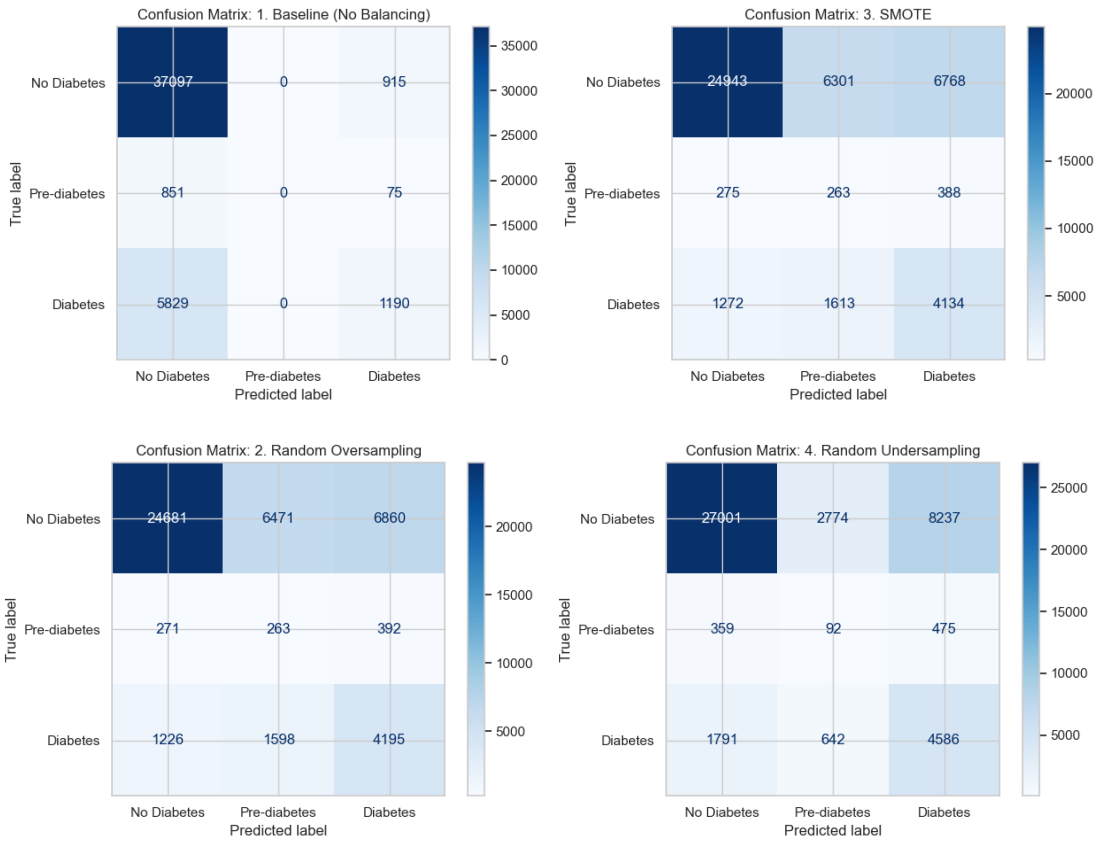

Figure 2: Softmax Confusion Matrices: Baseline (top-left), SMOTE (top-right), Oversampling (bottom-left), Undersampling (bottom-right).

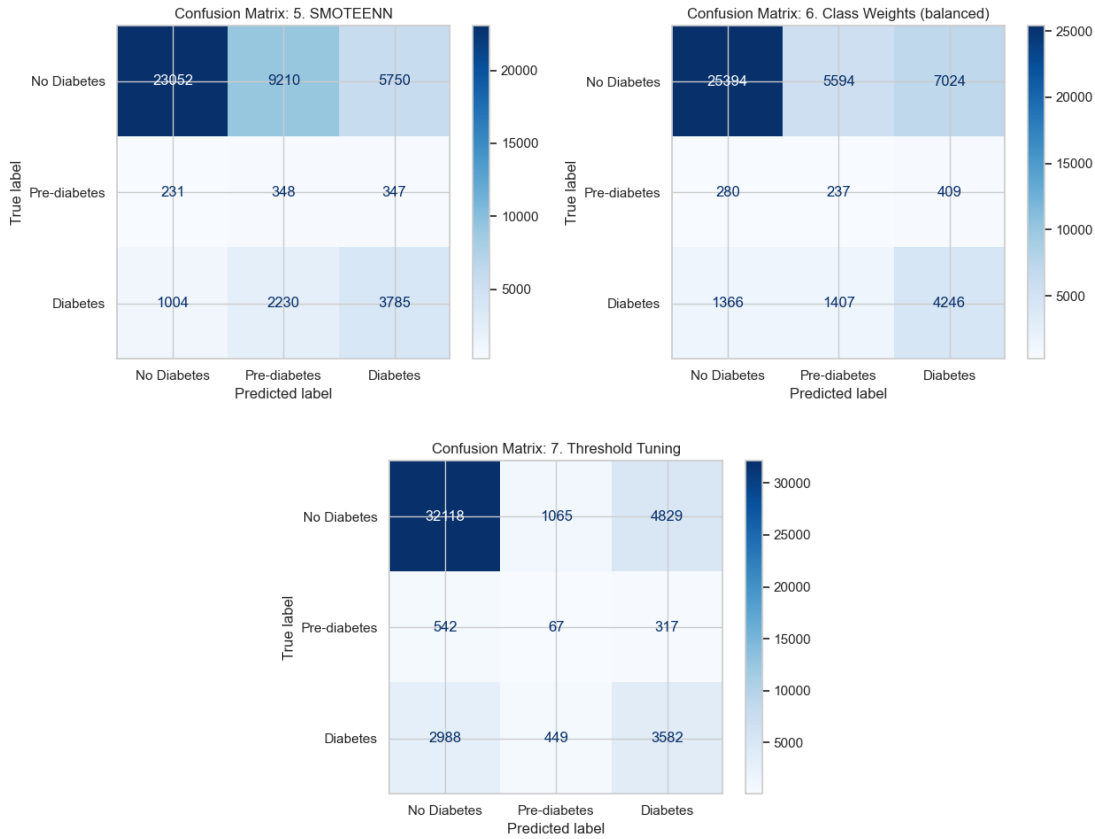

Figure 3: Softmax Confusion Matrices: SMOTEENN (top-left), Class Weights (top-right), Threshold Tuning (bottom-left).

3.3.3 Feedforward Neural Network (FNN)

Figure 4: FNN Confusion Matrices (Top Left: Baseline, Top Right: SMOTE, Bottom Left: Oversampled, Bottom Right: Undersampled)

4 Analysis and Comparison

4.1 Model Comparison

Among the models evaluated so far, the Softmax Regression and KNN models show a clear trade-off between raw accuracy and Macro-F1. The Softmax Baseline achieves the highest accuracy (83.31%) but a low Macro-F1 (0.3886), essentially predicting only the majority class. Once imbalance handling is applied, Softmax with Threshold Tuning achieves the best Macro-F1 among all Softmax variants (0.4602), outperforming SMOTE-based KNN (0.4166). This demonstrates that a linear model with proper calibration can exceed a distance-based model on this task.

The key insight is that Softmax Regression, being a linear model, is faster to train and tune across many strategies, while KNN is more sensitive to the local geometry of the feature space. Both models struggle significantly with the "Prediabetes" class due to its extreme rarity (only 2% of samples). Threshold Tuning (best thresholds: [0.1, 0.3,

0.3]) partially addresses this by aggressively lowering the decision boundary for minority classes.

4.2 Impact of Imbalance Handling

Comparing the baseline models to the balanced models clearly demonstrates the necessity of handling class imbalance. For KNN, the baseline model achieved a high overall accuracy of 81.66%, but suffered from a low Macro-F1 score of 0.3994, indicating it was heavily biased toward the majority class. Applying SMOTE reduced the overall accuracy to 71.77% but successfully increased the Macro-F1 score to 0.4166. This shift proves that the balancing technique improved the model's ability to predict the difficult minority classes (like Prediabetes), making the classifier much more clinically useful despite the mathematical drop in raw accuracy.

4.3 Insights and Misclassification Patterns

Inspecting the confusion matrices for Softmax Regression reveals a consistent and clinically important pattern: the **"Prediabetes" class is almost entirely misclassified as "No Diabetes"** across all strategies. Even with aggressive balancing (SMOTE, Undersampling, Threshold Tuning), the model rarely correctly identifies a Prediabetes sample.

This occurs for two compounding reasons:

1. **Mathematical:** The Prediabetes class represents only $\sim$2% of the dataset. Even after balancing the training set, the decision boundary learned by a linear model is dominated by the much larger "No Diabetes" and "Diabetes" clusters. The Prediabetes feature distribution overlaps heavily with "No Diabetes", making it nearly impossible for a linear separator to isolate.

2. **Clinical:** Prediabetes is, by definition, a borderline condition. Patients in this category share many health indicators with healthy individuals (normal BMI, moderate blood pressure), making the distinction genuinely ambiguous even for medical professionals. The features available in this dataset (binary lifestyle indicators) may not capture the subtle biochemical differences that distinguish Prediabetes from No Diabetes.

The Threshold Tuning strategy partially addresses this by lowering the decision threshold for the Prediabetes class, allowing the model to "bet" on Prediabetes more aggressively. However, this comes at the cost of increased false positives for that class.

5 Bonus: TensorBoard Monitoring

Figure 5: TensorBoard Training and Validation Loss Curves